

Percentage Change

AQA · KS3 Year 7

Name: _____ Class: _____ Date: _____

Calculate percentage increase and decrease. Find the percentage change between two values. Solve real-life problems involving percentage profit, loss and change.

VISUAL EXAMPLE



$$\frac{1}{2} = 0.5 = 50\%$$

shaded part = 1 of 2 equal parts = half

One half shown three equivalent ways.

COMMON MISTAKES

- ✗ Dividing by the NEW value instead of the ORIGINAL when calculating percentage change.
- ✗ Adding/subtracting the percentage of the new amount instead of the original.
- ✗ Confusing percentage increase with the new total — an increase of 20% means the new value is 120%.
- ✗ Forgetting to check if the change is an increase or a decrease.

METHOD

- 1 Find the percentage of the original: $(\text{percentage} \div 100) \times \text{original}$.
- 2 Percentage INCREASE: add the amount to the original.
- 3 Percentage DECREASE: subtract the amount from the original.
- 4 Percentage CHANGE: $(\text{actual change} \div \text{original}) \times 100$.
- 5 Check — state if the change is an increase or decrease.

WORKED EXAMPLE

A coat costs £80. It is reduced by 15%. Find the sale price.

Step 1 — Find 15% of £80: $15 \div 100 \times 80 = £12$

Step 2 — It's a decrease: $£80 - £12 = £68$

Answer: The sale price is £68.

1 Find:

(2 marks)

- a) 10% of £50 b) 25% of £120 c) 5% of £200 d) 1% of £350

HOW TO START

- To find 10%, divide the amount by 10.
- To find 1%, divide the amount by 100.
- 25% is a quarter, so divide by 4; 5% is half of 10%.

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change
= (change ÷ original) × 100 | 5 State increase or decrease

2 Increase each amount by the given percentage.

(2 marks)

a) £40 by 10% b) £60 by 25% c) £200 by 50%

HOW TO START

- First find the percentage of the amount (just like Q1).
- Then ADD that amount on to the original.

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

(2 marks)

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = (change ÷ original) × 100 | 5 State increase or decrease

(3 marks)

(a) = _____

(b) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = (change ÷ original) × 100 | 5 State increase or decrease

5 A bicycle cost £320. In a sale there is 15% off.

(3 marks)

- a) How much is the discount?
- b) What is the sale price?
- c) After the sale, the price goes back up by 15%. Is it £320 again? Show why or why not.

ANSWERS

- (a)** = _____
- (b)** = _____
- (c)** = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

(3 marks)

Answer = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = (change ÷ original) × 100 | 5 State increase or decrease

- 7** Mo buys trainers for £60 and sells them for £78.
Abi buys a bag for £45 and sells it for £54.

(4 marks)

- a) Calculate Mo's percentage profit.
- b) Calculate Abi's percentage profit.
- c) Who made the better deal? Explain.

ANSWERS

- (a)** = _____
- (b)** = _____
- (c)** = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change
= (change ÷ original) × 100 | 5 State increase or decrease

8 A house was worth £200,000. Its value increased by 12% in Year 1, then (5 marks)
decreased by 8% in Year 2.

- a) What was the value after Year 1?
- b) What was the value after Year 2?
- c) What is the overall percentage change from the original price? Is it an increase or decrease?

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = (change ÷ original) × 100 | 5 State increase or decrease

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet

☐ Getting there

☐ Confident

☐ Really confident

Which question did you find the hardest, and why?

One method step I want to remember next time:

One thing I will practise before the next lesson:

Teacher key

Q1: a) £5 b) £30 c) £10 d) £3.50

Q2: a) £44 b) £75 c) £300

Q3: a) £64 b) £135 c) £325

Q4: a) £45 b) $(45 \div 450) \times 100 = 10\%$

Q5: a) $15\% \text{ of } 320 = £48$ b) $320 - 48 = £272$ c) $15\% \text{ of } 272 = £40.80$; $272 + 40.80 = £312.80 \neq £320$. No — the 15% increase is on a smaller amount.

Q6: Change = 50; $(50 \div 250) \times 100 = 20\%$ increase

Q7: a) $(18 \div 60) \times 100 = 30\%$ b) $(9 \div 45) \times 100 = 20\%$ c) Mo — higher percentage profit

Q8: a) $200,000 \times 1.12 = £224,000$ b) $224,000 \times 0.92 = £206,080$ c) Change = £6,080; $(6,080 \div 200,000) \times 100 = 3.04\%$ increase
